ModularRSI: Modular and Generalizable Recursive Harness Self-Improvement

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HarnessRSI 72 ★★★

evidence: retained

Paper

Code

1 What changes, what is kept

Harness code split into 5 modules, evolved separately

Agent Loop · Observation · Tool Use · Context · Task Completion

1 Contrast pass vs fail
3 rollouts per task; compare the passing and failing run of a task

2 Vote, then patch one module
edits ranked by # of tasks backing them (top K = 10); one module only

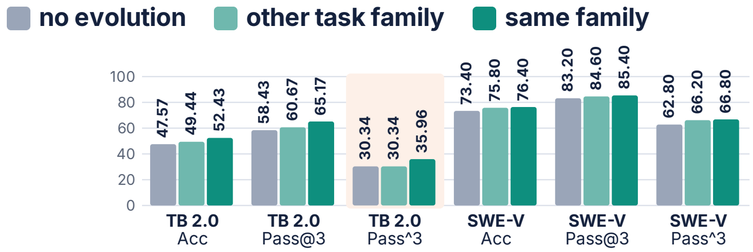
3× epochs

3 Three gates, else roll back
program check · diff review for task-specific hacks · run 2 tasks

4 Integrate once, freeze
1 cross-module epoch (16 in all); test time: an LLM composer picks the functions used for each task

two separate 120-task evolution sets, each gives its own harness; weights never change

2 Headline: TB 2.0 accuracy 47.57 → 52.43



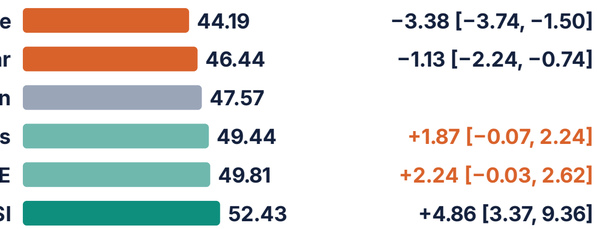
DeepSeek-V4-Flash-Preview, 3 rollouts per task, one harness per bar (Table 1). Same family: TB-related set → TB 2.0, SWE-related → SWE-V (500 tasks). Shaded: Pass^3 unchanged for the other-family harness.

4 Where the gain is absent or thin

- **Other optimizers are close:** 49.44 / 49.81 vs 52.43; 7‡ of 267 rollouts over AHE (‡ computed by us), no direct test
- **Accuracy-specific:** joint and non-modular beat no evolution on Pass@3 (61.80, 64.04 vs 58.43); non-modular needs fewer steps (29.03 vs 35.57)
- Other-family harness: TB 2.0 **Pass^3 stays 30.34**
- **SWE-V headline 76.40** = the better of two difficulty variants (Table 10; 74.20 for the other); selection not stated
- **One evolution run per cell;** CIs resample tasks only. Evolution curve (Fig. 6) runs to generation 22, not 16, and dips near 12
- "Matched" = 16 epochs, no cost figures. Web search off "for all methods" in Table 6, unstated for Table 1. Leakage screen unquantified

3 Controls that decide if the gain is real

TB 2.0 accuracy, 16 matched epochs · Δ vs none [95% CI]



Tables 3, 6, 9; CIs resample tasks only (orange: includes 0). Baselines re-run by the authors; tuning budget not reported; **not stated whether they got ModularRSI's composer, merge or gates.**

Ablation (Table 5, single runs, no CIs)

Library management = function merge + an LLM composer at test time; its model and cost are not reported.

Transfer (Tables 2, 9)

ModularRSI harness, accuracy	Δ [95% CI]
SWE-V, SWE-related evolution	+3.00 [1.06, 4.26]
SWE-V, TB-related (other family)	+2.40 [0.73, 3.33]
TB 2.0, SWE-related (other family)	+1.87 [1.49, 5.24]
TB 2.0, GLM-5.2 (frozen harness)	+2.25 [1.12, 5.24]
TB 2.0, MiniMax-2.5 (frozen harness)	+3.37 [2.25, 7.49]

Takeaway

Evolving five modules separately from contrastive evidence, behind rollback gates, beat joint and unrestricted evolution on accuracy at 16 matched epochs and transferred across task family and model. The margin over Meta-Harness and AHE has no direct test; every figure is one evolution run.

Our read · HarnessRSI

Score 72 · ★★★ · evidence level: retained

Reports: disjoint data, transfer, matched-epoch baselines, task CIs.
Does not report: run-to-run variance, cost, ablation CIs, baselines' composer/gates, quantified leakage check.